

GREAT SCIENTISTS

Roger Bacon

The 13th-century English friar Roger Bacon (c 1214–1292) was nicknamed “Doctor Mirabilis” (Wonder Doctor) for his wide-ranging scientific interests. At a time when universities taught very few subjects in an unchanging curriculum, Bacon wanted to introduce a different type of education.

Bacon the monk

Bacon studied at the University of Oxford in the 1230s, before moving to Paris as a lecturer. He returned to Oxford in 1247, where he began his scientific research. In 1257 he became a Franciscan monk. Living in a strict religious community, he found it hard to continue his experimental work.

Reform of universities

Medieval university students learned mainly theology (study of religious belief) and were also introduced to grammar, logic, and rhetoric (the art of speaking and writing effectively). Classical Greek and Latin authors such as Aristotle were used as models. In his important book *Opus Maius* (Latin for “Greater Work”), Bacon argued for a much wider range of subjects, including optics, geography, mechanics, and alchemy (medieval chemistry).

Optics

Bacon had new ideas about vision. While Greek scientists believed sight was caused by a ray that came from the eye, Bacon thought all objects gave out a wave that rippled outward. When this wave reached the eye, the object was seen.

Later years

In 1268, Bacon lost a protector when Pope Clement IV, who supported Bacon’s work, died. Many Franciscans thought Bacon’s ideas went against the teachings of the Catholic Church. He may even have been imprisoned for a while in Italy. Bacon eventually returned to England where he wrote new works, including one on mathematics. He died in 1292.

Opus Maius

This drawing of an eye is taken from Bacon’s book *Opus Maius*. Bacon was not allowed to publish without the consent of the Franciscans, but in 1266 Pope Clement IV asked him to produce a summary of all the things he believed should be taught in universities. By the time Bacon had completed his work in 1268, Clement had died.

Bacon holds scales to measure the weight of substances.

Scale pan containing the element water

Scale pan containing the element fire in balance with left-hand pan

The alchemist

Like many scholars of the time, Bacon practiced alchemy. Alchemists believed that everything was made of four “elements”: earth, air, fire, and water. They also thought they could transmute (transform) metals such as lead into gold.



Lecturer in Paris

Bacon is seen here presenting one of his works to the Chancellor of Paris University. Bacon taught there for 10 years from around 1240. He met other scholars, such as Peter Peregrinus, who wrote a work on magnetism and inspired Bacon’s love of experimenting.